The TAKOMO operating model

What goes in, what happens on the floor, who builders get connected to and what comes out. The exact function.

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0. The shape in one line

TAKOMO takes builders, improves their skills with hard, physical technology, then connects each one to wherever they actually want to go. That is the core idea. People arrive with a skill, an idea or just curiosity; and they leave more capable having gained the knowledge, the hands-on time and connections to the people they could never reach on their own. What comes out of the place is skilled people, shipped projects and open source products. These bring along new tech and potentially patents which a company can buy and every so often, a funded startup. All of it runs on shared industrial equipment and shared expertise that none of these people could afford alone.

There are two easy ways to misread this. One is the hobby room full of tools, where people make small things and go home. The other is the startup machine, where an idea goes in and a company comes out. Both miss the same point and the whole model turns on getting three things right.

1. The core of it is capability, not companies. The main thing TAKOMO gives a builder is the ability to do deep, hard technology they could not manage alone. The equipment matters, but the knowledge and the people who already know how matter more. Everything else the place produces follows from making people more capable. Lend out machines and nothing else and you have a tool library, which we do not limit ourselves to.

2. Builders head in many different directions and starting a company is only one of them. Some are here to learn and build for its own sake and stay for good. Some want to work on the real problems partners bring in. Some want to go deep into the science. Some want a job, a product to ship, or a patent to sell. Each of these is a real result and each matters to a different one of the groups funding the place. Treating a startup as the destination throws most of that away.

3. The engine is the connection layer, not the room. Making people capable is half of it and matching is the other half. Connecting the right builder to the right person, problem, lab, or partner is what turns that capability into somewhere to go and it decides which of the many directions above a builder actually reaches.

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1. The five-sided exchange

This is who puts in what and who gets what back. The entire model exists to keep all five of these columns full at the same time. If any one of them runs dry, whether that's the venture capital, the corporate problems, the grant and donor funding, the research indirectly coming through or the builders themselves, the line stops moving.

Side	Puts in	Takes out	Harvests at
Builders / members	Time, ideas, energy, skill	Skills, network, projects, a company or a job	Throughout
Investors / VCs	Capital, access to dealflow	Fundable startups, equity, a curated pipeline	When a company spins out
Corporates	Real problems, sponsorship, pilots, first-customer demand	Validated tech, pilots, acquisitions, patents, a hiring pipeline	From applied work on
Donors and foundations	Grants, donations, in-kind support and a mission to back	Mission impact they can report: talent trained, capability built, things seeded in the region	Throughout
Aalto	Talent, facilities, credibility	Visibility, spinouts, engaged students	Throughout

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2. Inputs: what goes in

People

- **Members:** students, researchers, engineers between jobs, dropouts, tinkerers and hobbyists. Sort them by how far they actually want to go (just curious, hobbyist, building a real project, trying to found a company), because that tells you more about their path than their skill level does.
- **Founding teams,** either already formed or put together inside the space.
- **Corporate engineers** embedded on partner projects.

Problems and ideas (the fuel that feeds everything else)

- Ideas that members bring in themselves.

- **Corporate briefs.** Partners post real problems as open challenges. This matters because it gets corporate demand in at the front of the process, not just at the exit.
- Gaps in the market that mentors and partners spot.

Infrastructure

- The shared bench: equipment across a spread of domains (electronics, RF, mechanical and CNC, fabrication, prototyping, test), cleanroom-adjacent access, safety systems and the technical staff who actually run the machines.

Capital

- Funds and foundations, corporate sponsorship and prize or challenge pools.

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3. The core: the branching paths

A builder doesn't ride a single conveyor from the front door to a startup and most of them never want to. TAKOMO is first of all a place to build, learn and work on real things alongside other people. That is the main event, not a waiting room before something more serious. Some of that work grows into a project with a partner, some into research and once in a while into a company, but those are directions a few paths lead to, not the reason the place exists.

Everyone comes in through the same short trunk, spends most of their time on the shared floor and can head off in different directions from there. Almost all of it loops back to the base.

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Entry
└─ The Commons: build, learn, use the shops, be part of it
   │  (the main thing and where most members stay for good)
   ├─ Going deeper: harder builds, shipping projects, teaching others
   ├─ Applied projects: real problems that partners bring in
   ├─ Research: deep science with an Aalto lab
   └─ Starting a company: the exception, with the accelerator there for it
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The trunk: where most of it happens

Entry. Someone joins, gets safety certified, learns the basic tools and has their profile recorded: skills, interests and what they're here for. Once they're certified, they have floor access.

The Commons. The shared floor and the community around it and the main thing TAKOMO is. People learn tools, use the shops (robotics, metal, studio and the general workspace), build their own projects and help on other people's, run and take workshops and spend time in the lounge. Most members live here and never need anything else and that is the whole point, not a holding pattern. It is where skills spread, where people meet the collaborators they end up working with and where the place gets its energy. Protect it. Everything else in this section is smaller than this.

Going deeper. Inside the Commons, people grow from a newcomer to a solid builder to someone who helps and then teaches the next wave. There's no gate and no finish line and staying to build and mentor is one of the most valuable things a member can do. It produces shipped projects, open-source work and the mentors who keep the floor alive.

What some of the work grows into

Three directions a project or a builder can also take. None is required and most work never leaves the Commons.

Applied projects are for people who want to work on the real problems partners bring in, whether or not anything more comes of it. They pick up a brief from the board, scope it with the partner, build and pilot it on the bench and deliver. Partners get tech they can validate and sometimes buy and members get real-world problems and, often, a job offer. If someone decides they want to own the work rather than hand it off, that is where the company path opens up.

Research is for people who want to go deep on the science. They pair with an Aalto lab or a supervisor, run the real investigation and produce results worth publishing or protecting. It leads to papers, IP, PhD outcomes and visibility for Aalto. This is where the Act 369/2006 ownership question from section 2 bites hardest, so that intake rule has to be settled before this produces anything valuable.

Starting a company happens now and then, not as the goal but as one thing a strong project can turn into. When a build has real pull behind it, a working version and someone who wants it, the attached accelerator steps in with funding, mentorship and help closing a first customer and the deal desk handles incorporation and IP. Most projects never go this way and they don't need to. Naming it is only so the few who do have a clear road, not so it becomes the road for everyone.

Moving between branches

None of these are sealed lanes. A builder who stumbles onto something worth selling can take it toward a company. An applied project someone wants to own becomes their own thing. A company that doesn't get off the ground turns its result into a paper, or the person drops back to the Commons and helps the next group through. The one constant is the trunk: everyone comes in the same way and can always come back to the floor, which is what makes moving between directions possible at all.

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4. The connection layer: who builders get connected to and how

This is the real engine of the place. The space lets people build things; this layer is what decides whether those builds turn into output. For each kind of connection, there's who or what sets it off and how it actually happens in practice.

Connect builder to	Why	How
Peers and co-founders	Teams, complementary skills	Roster and profiles, matching sessions, hack events
Technical staff and equipment	Turn a sketch into a build	Induction, help on the floor, booking
Mentors and experts-in-residence	Unblocking, technical and business	Set office hours, residencies
Corporate partners	A problem, a pilot, a first customer, a buyer, an employer	Brief board, demo days, matched pilots
Capital	Funding the company	Accelerator intake, angel and VC demo days, grant support
Aalto research	Deeper science, IP, PhD routes	Lab access, joint projects, links to supervisors
Alumni and returners	Mentorship, angel cheques, credibility	Alumni network, returning-founder program

The routing doesn't happen by luck and someone owns it. There's a Community Lead, call them the connector, whose actual job is to take a builder who needs something and hook them up with whoever can provide it. Sitting under that person, four standing mechanisms do most of the heavy lifting.

- **The brief board.** Corporates post problems and builders opt into the ones they want. This is what gets corporate demand in at the top of the funnel instead of only at the end.
- **A regular demo cycle.** Showcases on a set rhythm where projects meet capital, corporates and Aalto in the same room. Most of the matches and most of the exits start here.
- **The review points.** When a project is ready to go further, toward a partner, a lab, or the accelerator, it gets a review and that review is also where it gets pointed at the right people.

- **The deal desk.** At the exit, it structures the deal, keeps the builder's options open between raising money and getting acquired and sorts the IP out cleanly.

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5. Outputs: the ways out

These are all the ways a project or a person can leave the building, or stay and go round again. I've split them by whether someone stays inside the loop or heads out of it.

Staying inside the loop (people doing more inside rather than leaving, which is what keeps the place full)

Exit	What it is	Who it's for
Recirculate	Stay, level up, start something again	TAKOMO (keeps it busy), the member
Returner	Comes back as a mentor or investor	TAKOMO (the flywheel)
Graceful fail	Didn't work, so learnings and a way back in	The member (who stays), TAKOMO

Going out (creating things outside the building, startups and everything else)

Exit	What it is	Who it's for
Startup spinout	A VC-fundable company leaves	Investors, the member
Acquisition / acqui-hire	A corporate buys the small company or the team	The corporate, the member
Patent license or sale	IP sold or licensed to a partner	The corporate, the member, Aalto
Open source output	A paper, IP, or documentation	The general public
Product	Something that exists out in the world	The member
Talent placement	A proven engineer gets hired	The corporate, the member

Two tensions here that you have to actively manage rather than pretend away.

First, raising money versus getting bought. A corporate that wants to acquire and a VC that wants to fund can end up circling the same project. TAKOMO's job is to stay out of it and keep the builder's options open, not to nudge them one way. Decide that this is the rule now, before it ever comes up live.

Second, IP at the exit. Whatever you decide about ownership when people join has to hold up against Aalto's rules and against a real acquisition. Settle who owns what when people join, not when there's a deal on the table.

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6. The flywheel

The whole thing feeds itself. More people building means more chance encounters, which means more projects and better ones. More projects mean more exits. Those exits bring back founders as mentors and build the place's reputation, which pulls in more talent, more capital and more corporate partners, which puts more people on the floor building again. Around it goes.

Every exit that leaves on good terms feeds the next turn of that wheel: a founder who made it work, a paper that got cited, an engineer who got hired, a corporate that walked out with a pilot that actually held up. Each one is a reason for the next intake to show up. But it only compounds if those people stay in touch after they've gone, which is exactly why the returning founders and the alumni network aren't a nice extra bolted on the side. They're what holds the loop together.

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7. The moat

Every individual piece of this already exists somewhere else. A makerspace has the equipment but can't fund you or sell for you. An accelerator has the capital but can't help you prototype. A university has the research but no machine for turning it into companies. A corporation has the demand but can't run an open community of builders.

What TAKOMO has that none of them do is all of it inside one loop: the bench, the capital, the corporate demand, the research and the community, together. That's what lets a single builder go the whole way, from a sketch to a shipped product or a published result and for the few who take it there, to a company someone funds or buys, without ever leaving the building or having to line up five separate relationships on their own.

The loop is the actual product. The thousand square metres is just the place it runs.